1.0 PURPOSE

To outline management policy for

- Operational planning and control
- Requirements for products and services
- Design and development of products and services
- Control of externally provided processes, products and services
- Production and service provision
- Release of products and services
- Control of nonconforming output

2.0 SCOPE

Applicable to all products & processes at NIHVA

3.0 DETAILS

3.1 Operational planning and control

3.1.1 The planning of controls is required to ensure consistent acceptability of products and services. Planning processes include the definition of quality objectives, development for required processes, establishment for appropriate verification programs and the requirement for records necessary to demonstrate the process and products conform to intended requirements. NIHVA plans, implements and controls the processes needed to meet the requirements for the provision of products and services, and to implement the actions determined in Planning by:

- Determining the requirements for the products and services
- Establishing criteria for the processes and the acceptance of products and services
- Determining the resources needed to achieve conformity to the product and service requirements
- Implementing control of the processes
- Determining, maintaining and retaining documented information to the extent necessary to have confidence that the processes have been carried out as planned, and to demonstrate the conformity of products and services to their requirements

The output of operational planning and control includes documented customer inputs, resource requirements, processes, equipment requirements, procedures, installation/ Project data, and design outputs.

Operational Planning and Control



3.1.2 Quality planning includes following activities: -

- Technology, equipment, processes & skills required to achieve desired quality are identified and acquired.
- Compatibility of design, production processes, inspection & test procedures and related documentation is ensured.
- Quality records are identified, prepared and maintained.
- The updating, as necessary of quality control, inspection and testing techniques, including the development of new instrumentation.
- The identification of any measurement requirement.
- The identification of suitable verification at appropriate stages in the realization of product
- The clarification of standards of acceptability for all features and requirements, including those that contain a subjective element.

3.1.3 Control over outsourced processes: - In case NIHVA chooses to outsource any process that affects product conformity with requirements, it is ensured that appropriate control is exercised over such processes. The type and extent of control to be applied to outsourced process can be influenced by factors such as

- The potential impact of outsourced process on NIHVA's capability to provide product that conforms to requirements
- The degree to which the control for the process is shared
- The capability of achieving the necessary control

The outsourced processes at NIHVA for its QMS are –

- Calibration (by outside agencies)
- Subcontracted activities for commissioning

The controls exercised on these processes are:

- Calibration: - Ensure traceability.
- Subcontracted activities – adequate controls are exercised through supervision by NIHVA

3.2 Requirements for products and services

3.2.1 Customer Communication includes,

- Providing information related with products and services
- Handling enquiries, contracts or orders including changes
- Obtaining customer feedback related with products and services, including customer complaints
- Handling or controlling customer property
- Establishing specific requirements for contingency actions, when relevant.

3.2.2 Customer communication: - Procedures and Responsibilities are established within marketing department for effective communication with customer in relation to product information, inquiry, contract or order handling, order amendments, customer feedback including complaints. The primary means of communication are defined in the referenced procedures; however, at any time other ad hoc methods may be used as appropriate for a given business situation.

3.2.3 Determining the requirements for products and services: NIHVA has established processes to ensure that the requirements for products and services are defined.

Following requirements are determined:

- Requirements specified by the customer, including delivery and post-delivery activities. The post-delivery activity include actions under warranty provisions, contractual obligations such as maintenance services, and supplementary services such as recycling or final disposal.
- Requirements not stated by the customer but necessary for specified use or known intended use;
- claims for products and services
- Statutory and regulatory requirements applicable to the product; and
- Any additional requirements considered as necessary by NIHVA.

3.2.4 Review of Requirements Related to Products and Services

NIHVA Technologies ensures we have the ability to meet the requirements for products and services to be offered to customers. Management conducts a contract/product review prior to committing to supply products and services to a customer. The review process at a minimum includes:

- requirements specified by the customer, including the requirements for delivery and post- delivery activities;
- requirements not stated by the customer, but necessary for the intended use, when known;
- requirements specified by the organization;
- statutory and regulatory requirements applicable to the products and services
- Contract or order requirements differing from those previously expressed.

NIHVA Technologies ensures contracts, purchase orders or other requirements differing from those previously defined, are reviewed and approved prior to incorporating into our business systems.

We retain applicable documented information of the initial review and on any new/revised customer or applicable external party requirements for the products and services provided.

3.2.5 Changes to Requirements for Products and Services

NIHVA Technologies ensures that relevant documented information is amended, and that relevant persons are made aware of the changed requirements, when the requirements for products and services are changed.

3.3 Design and development of products and services

Design and Development of Products and Services are not applicable to traded items but the application of design and development will be as per specific customer requirement to meet the application, assembly, requirements in the field to meet the performance requirement.

Justification – NIHVA Technologies does not perform design activities for traded items since with basis of set vision and the new markets requirement, NIHVA planned to go ahead with developing in-house manufacturing capability including design to specific product application. NIHVA Technologies verifies the output of our customers design through measurements, fit checks, and visual inspections and suits to product application and necessary approval

3.4 Control of externally provided processes, products and services:

NIHVA Technologies maintains responsibility for the quality of all products purchased from external providers, including customer designated sources. Procedures ensure products and services being provided by external sources will conform to our customers' requirements. Examples of our controls include:

- A documented Approved Vendor List (AVL);
- The review of external provider's performance

3.4.1 Type and extent of control: NIHVA has established a Procedure to ensure that externally provided processes, products and services do not adversely affect the ability to consistently deliver conforming products and services to customers.

Purchasing procedures are established to

- Ensure that externally provided processes remain within the control of its QMS
- Define both the controls that it intends to apply to an external provider and those it intends to apply to the resulting output
- Take into consideration, the potential impact of the externally provided processes, products and services on the organization's ability to consistently meet customer and applicable statutory and regulatory requirements
- Take into consideration, the effectiveness of the controls applied by the external provider
- Determine the verification, or other activities necessary to ensure that the externally provided processes, products and services meet requirements.

Verification of purchased product: -

- Procedures are defined for receiving inspection activities.
- Where NIHVA or its customer intends to perform verification at the supplier's premises, the intended verification arrangements and method of product release are specified in the purchasing information.

3.4.2 Information for external providers: Purchasing processes are established to ensure the adequacy of requirements prior to their communication to the external provider.

Purchase order describes the product or service to be purchased including where applicable, following requirements.

- The processes, products and services to be provided
- Requirements for approval of products and services / methods, processes and equipment / release of product and services
- Requirements for competence, including any required qualification of persons
- The external providers' interactions
- Control and monitoring of external providers' performance to be applied
- Requirements about verification or validation activities
- Quality management system requirements.

3.5 Control of Production and service provision

NIHVA Technologies plans and implements production and service provision and as required by job specific requirements. Manufacturing/ Assembly procedures, AMC Contracts, inspection plans, and other documents deemed necessary, define the acceptance for manufacturing and service operations. The plans provide detailed instruction and guidance for all production and service phases

including the methods and equipment to be used. Records for each sales order of product produced/ install/ commission provide unique traceability. This record is maintained as required by customer contract requirements

3.5.1 A procedure is established to_plan and carry out production and service provision under controlled conditions. Controlled conditions includes the following as applicable

- The availability of documented information that defines the characteristics of the product to be produced, the services to be provided, or the activities to be performed and the results to be achieved
- The availability and use of suitable monitoring and measuring resources
- The implementation of monitoring and measurement activities at appropriate stages to verify that criteria for control of processes or outputs, and acceptance criteria for products and services, have been met
- The use of suitable infrastructure and environment for the operation of processes
- The appointment of competent persons including any required qualification
- The validation and periodic revalidation of the ability to achieve planned results of the processes for production and service provision, where the resulting output cannot be verified by subsequent monitoring or measurement
- The implementation of actions to prevent human error
- The implementation of release, delivery and post-delivery activities.

3.5.1.1 Procedure is established to validate any processes for production and service provision where the resulting output cannot be verified by subsequent monitoring or measurement. This includes any processes where deficiencies become apparent only after the product is in use or the service has been delivered. Following processes have been identified as special processes

- Welding
- Surface treatment processes like Plating

3.5.1.2 Arrangements are established for these processes including, as applicable

- Defined criteria for review and approval of the processes,
- Approval of equipment and qualification of personnel,
- Use of specific methods and procedures,
- Requirements for records
- Revalidation.

3.5.1.3 Operators are qualified by test or experience

3.5.2 Identification and traceability: -

- NIHVA Technologies identifies parts and products by suitable means for supplies to customer. According to the level of traceability required by contract, regulatory or other established requirement, our procedures provides for:
- identification to be maintained throughout the processes including delivery and post-delivery;
- Identification of sub-components and those of the next higher assembly.
- Procedures and responsibilities are defined for identification of the inspection status through all phases of production.
- Inspection and test status is identified either by tags or by storage locations that are clearly identified or by a document or on the material itself.
- When required by the customer, additional verification and identification requirements are met.

3.5.3 Property belonging to Customers or external providers

- In case of customer / external provider supplied property (parts or intellectual property); procedure and responsibilities are established for Receiving, Receiving inspection, Marking, Handling, Storage and Maintenance.
- Procedure is established to identify, verify, protect and safeguard customers' or external providers' property provided for use or incorporation into the products and services
- Any such property which is lost, damaged or otherwise unsuitable for use is recorded and reported to the customer / external provider
- Production tooling owned by customer / external provider (if available) are marked permanently

3.5.4 Preservation

- NIHVA Technologies preserves the conformity of parts and products during internal processing and delivery to the intended destination including outside services. Procedures include instructions for identification, handling, packaging, storage and protection. Preservation of outputs also includes, where applicable:
  - cleaning;
  - prevention, detection and removal of foreign objects;
  - special handling for sensitive outputs;
  - marking and labeling including safety warnings;
  - Special handling for hazardous materials.
- The shipping department ensures that documents required by the contract/order to accompany the product are present at delivery and are protected against loss and deterioration

3.5.5 Post-delivery activities: -

- While establishing procedure for post-delivery activities, NIHVA considers statutory and regulatory requirements, potential undesired consequences associated with its products and services, nature / use and intended lifetime of products and services, customer requirements and customer feedback.
- Responsibilities and procedures for product service are defined. Servicing activity is coordinated by service department

3.5.6 Control of Changes: -

NIHVA Technologies shall review and control changes to the extent necessary to ensure continuing conformity of customer or internal requirements. Changes may be initiated as a result of:

- needs of interested parties, or customer feedback;
- Opportunities for improvement are identified.

Management reviews and monitors changes that affect project or outside services and ensures change documentation and information is distributed and controlled. Records of results of the review of changes, the persons authorizing the change, and any necessary actions arising from the review are maintained in accordance with applicable procedures.

3.6 Release of products and services:

NIHVA Technologies monitors and measures the characteristics of the product in receiving inspection, in- process inspection, and final inspection to verify that requirements have been met. Documented

procedures have been established for product inspection. Documented Records and information of inspection include evidence of conformity with the acceptance criteria and traceability to the person authorizing the release. Records of inspection are maintained

3.7 Control of nonconforming outputs

- Procedures and responsibilities are established to ensure that outputs which does not conform to their requirements are identified and controlled to prevent its unintended use or delivery.
- A procedure is established to define the controls and related responsibilities and authorities for dealing with nonconforming output. Actions are initiated based on nature of the conformity and its effect on the conformity of products and services. This also applies to nonconforming products and services detected after delivery of products, during or after the provision of services
- Nonconforming products are identified and stored separately
- Nonconforming outputs are dealt with one or more of following ways
    - Correction
    - Segregation, containment, return or suspension of provision of products and services
    - Informing the customer
    - Obtaining authorization for acceptance under concession
- Conformity to the requirements are verified when nonconforming outputs are corrected
- Records of nonconformities are maintained as required and include:
    - description of nonconformity;
    - description of actions taken;
    - description of concessions obtained;
    - Identification of the authority deciding the action in respect of the nonconformity.
- When nonconforming product is corrected, it is re-inspected to the original specifications and requirements to ensure it conform to customer stated requirements. When a nonconforming product is detected after delivery, NIHVA Technologies will take action appropriate to the effects or potential effects of the nonconformity.

3.8 Related Documents

QSP-07-001 Control of Documents & Information	QSP-08-006 Vendor Evaluations
QSP-08-001 Control of Nonconformance's	QSP-08-007 Inspection
QSP-08-002 Engineering Change Notice	QSP-08-010 Preservation
QSP-08-003 Identification & Traceability	QSP-08-012 Contract Review
QSP-08-004 Rework	QSP-08-014 Shipping& Receiving
QSP-08-005Purchasing	QSP-08-015 Control of Customer Property